

Data Structure (C Programming)

25/08/2026

Name : Harshad Sanjay Teli.

Roll No : 15

Batch : B1

Prerequisite

Based on Structures.

Q1. Write a C program to define a structure **Student** containing name, roll number, and marks. Accept and display details of 5 students.

Input:

Assignment No 1: Bst-5 (Based on Structure)

Que1] Write a C Program to define a Structure Student Containing name, roll number and marks. Accept and display details of 5 Students.

```
#include <stdio.h>
struct Student {
    char name[20];
    int roll;
    float marks;
}

int main()
{
    struct Student s[5];
    int i;
    for(i=0; i<5; i++)
    {
        printf("\nEnter details of student %d\n", i+1);
        printf("Enter Name:");
        scanf("%s", s[i].name);
        printf("Enter roll number:");
        scanf("%d", &s[i].roll);
        printf("Enter Marks:");
        scanf("%f", &s[i].marks);
    }
    printf("\n... Student Details ... \n");
    for(i=0; i<5; i++)
    {
        printf("\n Student %d\n", i+1);
        printf("Name = %s\n", s[i].name);
        printf("Roll No = %d\n", s[i].roll);
        printf("Marks = %.2f\n", s[i].marks);
    }
    return 0;
}
```

Output:

```
C:\Users\Harshad Teli\Downloads\DS>output
Enter Details for the Student 1
Enter Name:Harshad_Teli
Enter Roll No:15
Enter Marks:98
Enter Details for the Student 2
Enter Name:Anurag_Lad
Enter Roll No:5
Enter Marks:99
Enter Details for the Student 3
Enter Name:Vardhan_Patki
Enter Roll No:68
Enter Marks:90
Enter Details for the Student 4
Enter Name:Shrinath_Teli
Enter Roll No:112
Enter Marks:95.68
Enter Details for the Student 5
Enter Name:Anjali_Patil
Enter Roll No:41
Enter Marks:86.44

-----STUDENT DETAILS-----

Name:Harshad_Teli
Roll No:15
Marks:98.00
Name:Anurag_Lad
Roll No:5
Marks:99.00
Name:Vardhan_Patki
Roll No:68
Marks:90.00
Name:Shrinath_Teli
Roll No:112
Marks:95.68
Name:Anjali_Patil
Roll No:41
Marks:86.44
C:\Users\Harshad Teli\Downloads\DS>|
```

Q2. Write a C program to add two distances (in feet and inches) using structures.

Input:

Q.2] Write a C program to add two distance (in feet and inches) using structures.

```
#include <stdio.h>
```

```
struct Distance {
```

```
    int feet;
```

```
    int inch;
```

```
};
```

```
int main() {
```

```
    struct Distance d1, d2, sum;
```

```
    printf("Enter 1st distance\n");
```

```
    printf("Enter feet: ");
```

```
    scanf("%d", &d1.feet);
```

```
    printf("Enter inch: ");
```

```
    scanf("%d", &d1.inch);
```

```
    printf("\nEnter 2nd distance\n");
```

```
    printf("Enter feet: ");
```

```
    scanf("%d", &d2.feet);
```

```
    printf("Enter inch: ");
```

```
    scanf("%d", &d2.inch);
```

```
    sum.feet = d1.feet + d2.feet;
```

```
    sum.inch = d1.inch + d2.inch;
```

```
    if (sum.inch >= 12) {
```

```
        sum.feet = sum.feet + sum.inch / 12;
```

```
        sum.inch = sum.inch % 12;
```

```
    }
```

```
    printf("\nSum of distances = %d feet %d inch\n", sum.feet, sum.inch);
```

```
    return 0;
```

```
}
```

Output:

```
C:\Users\Harshad Teli\Downloads\DS>output
Enter 1st distance
Enter feet: 5
Enter inch: 3

Enter 2nd distance
Enter feet: 6
Enter inch: 2

Sum of distances = 11 feet 5 inch

C:\Users\Harshad Teli\Downloads\DS>
```

Q3. Write a C program to store and display information of employees using an array of structures (fields: emp_id, name, salary).

Input:

Q.3] Write a C program to store and display information of employees using an array of structures (fields: emp_id, name, salary).

```
#include <stdio.h>
int n;
struct Employees
{
    int emp_id;
    char name[20];
    int salary;
}

int main()
{
    struct Employees e[100];
    int i, n;
    printf("How many Employees:");
    scanf("%d", &n);
    for(i=0; i<n; i++)
    {
        printf("Enter Employee %d details:\n", i+1);
        printf("Enter ID");
        scanf("%d", &e[i].emp_id);
        printf("Enter Name:");
        scanf("%s", &e[i].name);
        printf("Enter Salary:");
        scanf("%d", &e[i].salary);
    }
    for(i=0; i<n; i++)
    {
        printf("Employee %d Details\n", i+1);
        printf("Name: ID: %d\n", e[i].emp_id);
    }
}
```

```
printf("%d", e[i].name);  
printf("Salary: %d", e[i].Salary);  
}  
return 0;  
}
```

Name: Harshad Teli.

Roll No: 15.

Sign: Harshad Teli

Output:

```
C:\Users\Harshad Teli\Downloads\DS>output
```

```
How many Employees: 5
```

```
Enter Employee 1 details:
```

```
Enter ID: 1001
```

```
Enter Name: Anjali
```

```
Enter Salary: 45000
```

```
Enter Employee 2 details:
```

```
Enter ID: 1045
```

```
Enter Name: Smita
```

```
Enter Salary: 34000
```

```
Enter Employee 3 details:
```

```
Enter ID: 2026
```

```
Enter Name: Anurag
```

```
Enter Salary: 90000
```

```
Enter Employee 4 details:
```

```
Enter ID: 7045
```

```
Enter Name: Vardhan
```

```
Enter Salary: 45000
```

```
Enter Employee 5 details:
```

```
Enter ID: Siddhi
```

```
Enter Name: Enter Salary: 6767
```

```
Employee 1 Details
```

```
ID: 1001
```

```
Name: Anjali
```

```
Salary: 45000
```

```
Employee 2 Details
```

```
ID: 1045
```

```
Name: Smita
```

```
Salary: 34000
```

```
Employee 3 Details
```

```
ID: 2026
```

```
Name: Anurag
```

```
Salary: 90000
```

```
Employee 4 Details
```

```
ID: 7045
```

```
Name: Vardhan
```

```
Salary: 45000
```

```
Employee 5 Details
```

```
ID: 1
```

```
Name: Siddhi
```

```
Salary: 6767
```